

Camila Gonçalves dos Santos

M.Sc. Candidate in Remote Sensing

National Institute for Space Research (INPE)

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PROFILE

M.Sc. candidate in Remote Sensing at the National Institute for Space Research (INPE), with research experience in Earth observation, satellite image processing, environmental monitoring, and machine learning. My research focuses on developing methodologies using multi-source satellite imagery and time-series analysis to characterize environmental dynamics. I am particularly interested in satellite remote sensing, data fusion, and developing retrieval methods to improve the understanding of Earth system processes, including clouds, precipitation, and climate.

EDUCATION

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| 2025–Present | M.Sc. in Remote Sensing
<i>National Institute for Space Research (INPE) — São José dos Campos, SP, Brazil</i>
Advisor: Dr. Thales Sehn Körting
Research Topic: Remote sensing-based phenological characterization of degraded areas in the Triunfo do Xingu Environmental Protection Area using multi-sensor satellite time series, vegetation indices and machine learning. |
| 2018–2024 | B.Sc. in Geography (Licentiate and Bachelor)
<i>Federal Rural University of Rio de Janeiro (UFRRJ) — Seropédica, RJ, Brazil</i>
Advisor: Dr. Gustavo Mota de Sousa
Thesis: Multitemporal mapping and analysis of land cover and land use dynamics in the Western Metropolitan Region of Rio de Janeiro |

RESEARCH EXPERIENCE

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| 2025–Present | M.Sc. Researcher (CAPES Fellow)
<i>National Institute for Space Research (INPE)</i> <ul style="list-style-type: none">• Developing remote sensing methodologies using multi-sensor satellite time series.• Phenological characterization of degraded Amazonian ecosystems.• Application of machine learning techniques (Random Forest and XGBoost).• Processing CBERS-4, CBERS-4A and AMAZONIA-1 imagery. |
| 2024–2025 | Undergraduate Research Fellow (CNPq)
<i>National Institute for Space Research (INPE)</i> <ul style="list-style-type: none">• Rapid disaster mapping using Earth Observation satellite imagery.• Development of Python workflows for image processing.• Production of Value Added Products (VAPs).• Flood mapping using deep learning (ConvLSTM). |
| 2020–Present | Research Assistant
<i>Integrated Laboratory of Applied Physical Geography (LiGA), UFRRJ</i> <ul style="list-style-type: none">• Land use and land cover mapping.• Environmental monitoring using remote sensing and GIS.• Fire hotspot analysis.• Machine learning applied to environmental monitoring. |

TEACHING EXPERIENCE

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| 2023–2024 | Teaching Assistant (Scholarship)
<i>Remote Sensing Course — Federal Rural University of Rio de Janeiro</i> <ul style="list-style-type: none">• Assisted undergraduate laboratory classes in Remote Sensing.• Guided students in GIS and image processing activities.• Supported teaching, research and extension activities developed by LiGA. |
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PUBLICATIONS

- Santos, C.G., & Sousa, G.M. (2026). **Land Cover Mapping and Multitemporal Analysis of Land Use Dynamics in the Western Metropolitan Region of Rio de Janeiro.** *Revista Tamoios*. Accepted.
- Ojeda, A.P.W., Santos, C.G., Garcia Junior, C.P.G., Conceição, G.W., & Carvalho, L.P.C.F. (2024). **SAFE: Georeferencing of Bus Robberies in the State of Rio de Janeiro.** *FOCO Journal*.
- Santos, C.G., & Sousa, G.M. (2022). **Land Cover Mapping of the Western Metropolitan Region of Rio de Janeiro.** Proceedings of

SELECTED CONFERENCE PRESENTATIONS

ConvLSTM for flood mapping in Sentinel-2 time-series images. *Brazilian Congress on Risk and Disaster Reduction*, Florianópolis, Brazil, 2025.

Remote sensing for rapid disaster response. *INPE Scientific Initiation Seminar (SICINPE)*, São José dos Campos, Brazil, 2024.

Landscape analysis of the Western Metropolitan Area of Rio de Janeiro through land cover dynamics. *Seminar of the Biogeography Study Group*, UFRRJ, Brazil, 2023.

TECHNICAL SKILLS

Programming	Python, R
Remote Sensing	Google Earth Engine, QGIS, ArcGIS Pro, TerraView, GeoDMA, ENVI, SNAP
Machine Learning	Random Forest, XGBoost, Deep Learning (ConvLSTM)
Methods	Time-Series Analysis, Image Processing, Data Fusion, Vegetation Indices, Land Cover Classification

AWARDS & SCHOLARSHIPS

2025–Present	CAPES Master’s Scholarship <i>National Institute for Space Research (INPE)</i>
2024–Present	CNPq Undergraduate Research Fellowship <i>National Institute for Space Research (INPE)</i>
2023–2024	Teaching Assistant Scholarship <i>Federal Rural University of Rio de Janeiro</i>
2021–2022	CNPq Undergraduate Research Fellowship <i>Federal Rural University of Rio de Janeiro</i>
2020–2021	Undergraduate Research Fellowship <i>Federal Rural University of Rio de Janeiro</i>
2018–2020	PDAI Extension Scholarship <i>Federal Rural University of Rio de Janeiro</i>

LANGUAGES

Portuguese	Native speaker
English	Professional working proficiency (reading, writing, speaking)

PROFESSIONAL SERVICE

2024–Present	Volunteer Analyst (VA) <i>International Charter Space and Major Disasters</i>
2024	GIS and Data Analyst Intern <i>Semove — Federação das Empresas de Mobilidade do Estado do Rio de Janeiro</i> • Spatial analysis and georeferencing applied to urban mobility and public safety. • Contribution to the development of the SAFE project for mapping events in the state of Rio de Janeiro.
2022	Short Course Instructor <i>VIII Geography Academic Week — Federal Rural University of Rio de Janeiro</i> • Developed and taught the “Thematic Mapping” technical short course.
2020–Present	Research Member <i>Integrated Laboratory of Applied Physical Geography (LiGA), UFRRJ</i>